

# Jenkins Distributed Build Lab

## Objective

Understand Jenkins distributed architecture by deploying a controller and an SSH agent on separate EC2 instances and verifying job scheduling, workspace isolation, controller-agent communication, and troubleshooting.

## Architecture

GitHub -> Jenkins Controller (EC2 t3.small) --SSH--> Jenkins Agent (EC2 t3.small).

## Infrastructure

Controller: Ubuntu, Jenkins 2.555.3, OpenJDK, EC2 t3.small. Agent: Ubuntu, EC2 t3.small, SSH launch, remoting.jar.

## Learning Objectives

Controller vs Agent architecture, SSH agents, labels, workspaces, remoting.jar, Disk Space Monitor, distributed builds.

## Testing Performed

Verified SSH launch, controller-agent communication, controller workspace, agent workspace, label-based scheduling.

## Problem Encountered

Disk Space Monitor warned that /tmp had less than 1 GiB free on both controller and agent despite sufficient disk space.

## Root Cause

Ubuntu mounts /tmp as a tmpfs (~953 MiB). Jenkins monitored `java.io.tmpdir=/tmp` and compared it against a 1 GiB threshold, producing a permanent warning.

## Investigation

Verified `df -h /tmp` showed only 1% usage. Script Console confirmed `java.io.tmpdir=/tmp`.

## Controller Fix

Created `/var/lib/jenkins/tmp` and configured `-Djava.io.tmpdir=/var/lib/jenkins/tmp` via a systemd override. After correcting a typo and restarting Jenkins, the warning disappeared.

## Agent Fix

SSH Build Agents Plugin v3.1097 has no JVM Options field. Created a Java wrapper script adding `-Djava.io.tmpdir=/home/ubuntu/jenkins-agent/tmp` and configured JavaPath to use the wrapper. After reconnecting the agent, the warning disappeared.

## Lessons Learned

Controller and agent are independent JVMs. `java.io.tmpdir` is JVM-specific, not a Linux environment variable. Ubuntu tmpfs can trigger false Disk Space Monitor warnings. JavaPath wrappers are useful for injecting JVM startup options into SSH agents.

## Cleanup

Terminate EC2 instances after testing to minimize AWS costs. Preserve documentation to recreate the lab later.